$M/G/3/3$ Loss System with Weather-Dependent Service Times

Ram Prasad

MATH 4428

April 28, 2026

Setup

- Messages arrive according to a Poisson Process with rate

$$\lambda = 2 \text{ per hour.}$$

- There are 3 channels. If at least one channel is free, the message gets serviced immediately.
- If all three channels are busy, the message is lost. No queuing or wait time.
- Service time depends on the weather at the moment of arrival.
- Weather-dependent service distributions:

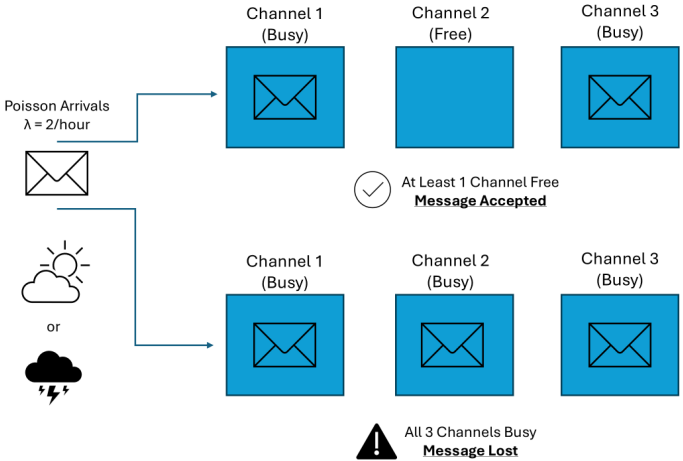
$$F_G(x) = x, \qquad 0 < x < 1,$$

$$F_B(x) = x^3, \qquad 0 < x < 1.$$

- Goal is to estimate the expected number of messages lost in the first 100 hours.

Mail System

M/G/3/3 Loss System with Weather-Dependent Service Times



Weather States: Bernoulli(p)

Bad Weather with prob. 1-p
Beta(3,1)
$F(x) = x^3$, $0 < x < 1$

Good Weather with prob. p
Uniform(0,1)
$F(x) = x$, $0 < x < 1$

Model:

- Message Arrives
- Realize Weather State
- Assign Service Time
- Enter Free Channel or Message Lost

Assumptions

- Sequence of events:

$$\text{arrival} \to \text{weather realization} \to \text{service time draw} \to \text{service or lost.}$$

- Arrival process (independent of service times):

$$M(t) \sim \mathrm{Poisson}(2t).$$

- Weather state at each arrival is i.i.d. Bernoulli:

$$W \sim \mathrm{Bernoulli}(p),$$

  where

$$\mathbb{P}(W = 1) = p, \qquad \mathbb{P}(W = 0) = 1 - p.$$

- $W = 1$: good weather, $W = 0$: bad weather.
- Conditional service time distributions:

$$X \mid W = 1 \sim \mathrm{Uniform}(0, 1), \qquad X \mid W = 0 \sim \mathrm{Beta}(3, 1).$$

Service Time Distribution

Conditioning on the weather state,

$$X = X_G \mathbf{1}_{\{W=1\}} + X_B \mathbf{1}_{\{W=0\}}.$$

Weather enters the loss probability calculation through the service time distribution, which is a two component mixture:

$$F_X(x) = pF_G(x) + (1-p)F_B(x), \qquad 0 < x < 1.$$

Substituting,

$$F_X(x) = px + (1-p)x^3, \qquad 0 < x < 1.$$

Differentiating gives the PDF,

$$f_X(x) = p + 3(1-p)x^2, \qquad 0 < x < 1.$$

Expected Service Time

By the law of total expectation,

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid W]\big] = p\,\mathbb{E}[X \mid W = 1] + (1-p)\,\mathbb{E}[X \mid W = 0].$$

The component first moments are

$$\mathbb{E}[X \mid W = 1] = \frac{1}{2}, \qquad \mathbb{E}[X \mid W = 0] = \frac{3}{4}.$$

Thus

$$\boxed{\mathbb{E}[X] = \frac{3-p}{4}}.$$

Similarly,

$$\boxed{\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{3}{80} + \frac{p}{120} - \frac{p^2}{16}}.$$

Erlang Insensitivity

For an $M/G/C/C$ loss system, the stationary blocking probability is *insensitive* to the shape of the service distribution and depends only on the mean service time.

Thus the $M/G/3/3$ system can be analyzed using the Erlang-B formula. Let

$$a = \lambda \mathbb{E}[X].$$

Using

$$\lambda = 2, \qquad \mathbb{E}[X] = \frac{3-p}{4},$$

the offered load (effective demand on the system) is

$$\boxed{a = 2 \cdot \frac{3-p}{4} = \frac{3-p}{2}}.$$

Loss Probability (Erlang-B)

Let p_k denote the steady-state probability of k busy channels. For an $M/G/3/3$ loss system,

$$p_k = \frac{a^k/k!}{\sum_{j=0}^{3} a^j/j!}, \qquad k = 0, 1, 2, 3,$$

with offered load $a = (3-p)/2$. A message is lost exactly when all 3 channels are busy, thus

$$\mathbb{P}(\text{message lost}) = p_3.$$

Using the Erlang-B formula,

$$p_3 = \frac{a^3/3!}{1 + a + a^2/2! + a^3/3!} = \frac{a^3/6}{1 + a + a^2/2 + a^3/6}.$$

Substituting $a = (3-p)/2$ gives

$$\boxed{p_3 = \frac{(3-p)^3}{201 - 87p + 15p^2 - p^3}.}$$

Expected Number of Messages Lost

Let $L_m(t)$ be the number of lost messages in the interval $[0, t]$.

Since the arrival process has rate $\lambda = 2$ and each arrival is blocked with probability p_3,

$$\mathbb{E}[L_m(t)] = \lambda t \cdot p_3 = 2t \cdot p_3.$$

Substituting and using $t = 100$,

$$\boxed{\mathbb{E}[L_m(t)] = 200 \left[\frac{(3-p)^3}{201 - 87p + 15p^2 - p^3} \right].}$$

I use this to compare to simulation results.

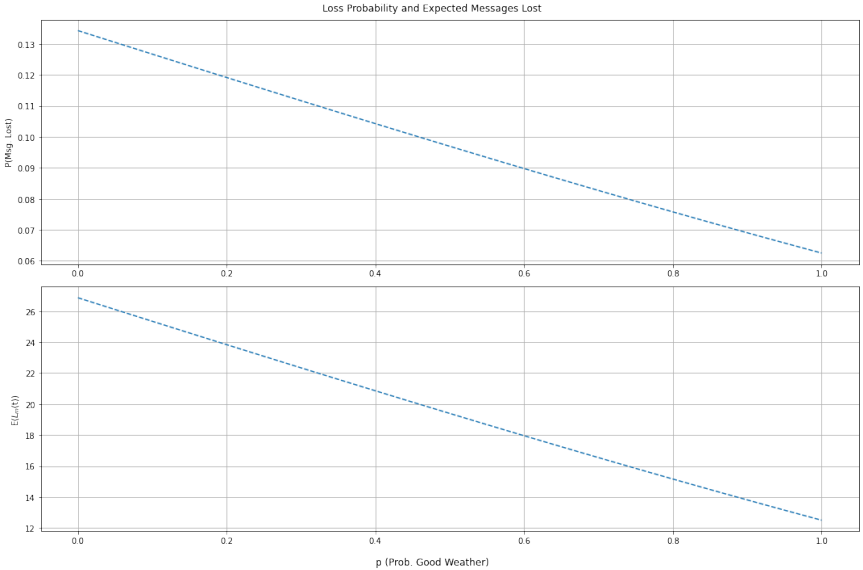
Key Cases

Case	$\mathbb{E}[X]$	p_3	$\mathbb{E}[L_m(100)]$
Always bad weather ($p = 0$)	$\frac{6}{8}$	≈ 0.134328	$200(0.134328) \approx 26.87$
Equal mixture ($p = 1/2$)	$\frac{5}{8}$	≈ 0.096974	$200(0.096974) \approx 19.39$
Always good weather ($p = 1$)	$\frac{4}{8}$	0.0625	$200 \cdot \frac{1}{16} = 12.5$

As p increases (good weather more likely)

- Expected service time falls
- Loss probability falls
- Expected number of messages lost falls

Loss Probability and Expected Messages Lost Across p



Simulation Exercise

To validate the analytic results, I simulate the process over $[0, 100]$.

1. Generate arrivals from PP(2) using exponential draws for interarrival times:

$$T_{i+1} - T_i \sim \mathrm{Exp}(2).$$

2. At each arrival time T_i, remove all messages that have been processed.
3. Draw weather state

$$W_i \sim \mathrm{Bernoulli}(p).$$

4. Draw service time

$$X_i = \begin{cases} U_i, & W_i = 1, \\ U_i^{1/3}, & W_i = 0, \end{cases} \qquad U_i \sim \mathrm{Uniform}(0, 1).$$

5. If fewer than 3 channels are busy, process the message and schedule departure at $T_i + X_i$.
6. If all 3 channels are busy, record a lost message.

Simulation Outputs

For each simulated path, the main statistics are the count of messages lost,

$$L_m(t) = \sum_{i=1}^{M(t)} L_i, \qquad L_i = \mathbf{1}\{\text{arrival } i \text{ is lost}\},$$

the fraction of messages lost,

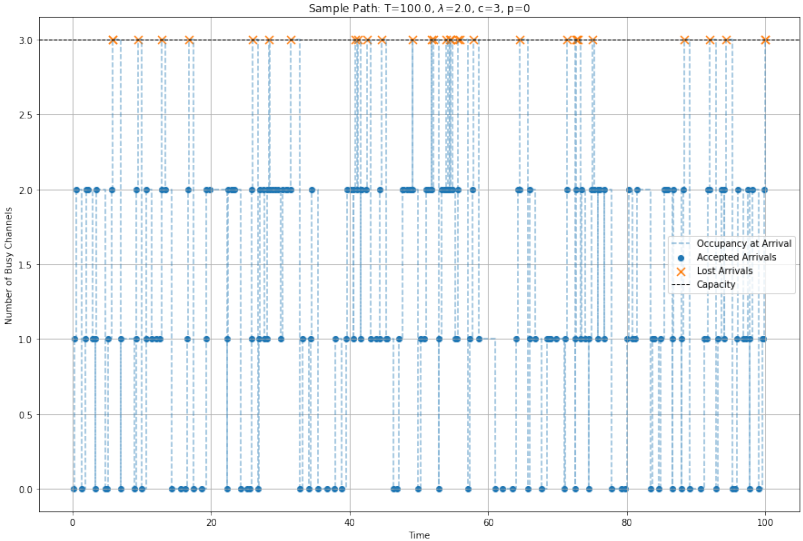
$$\widehat{p}_3 = \frac{1}{M(t)} \sum_{i=1}^{M(t)} L_i,$$

and the empirical mean service time over accepted jobs,

$$\widehat{\mathbb{E}[X]} = \frac{1}{N_{\text{accepted}}} \sum_{i=1}^{N_{\text{accepted}}} X_i.$$

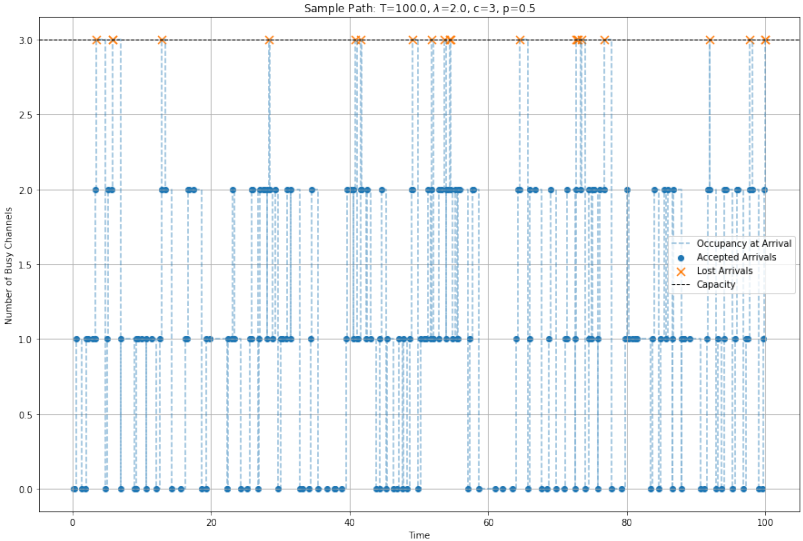
I compare the sampling distributions of the empirical quantities generated from $N = 1000$ runs against $\mathbb{E}[X]$, p_3, and $\mathbb{E}[L_m(100)]$.

Sample Path: Always Bad Weather ($p = 0$)



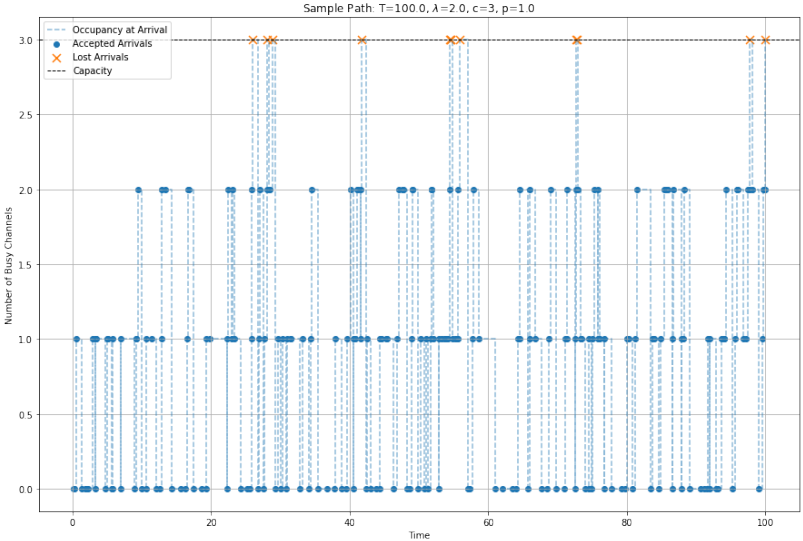
- Longer service times result in more congestion.
- System frequently operates near capacity ($c = 3$).
- Losses cluster during sustained busy periods.

Sample Path: Mixed Weather ($p = 0.5$)



- Shorter periods of sustained full capacity.
- Losses less frequent than in the $p = 0$ case.
- Higher probability of good weather results in lower offered load.

Sample Path: Always Good Weather ($p = 1$)



- Service times are shortest on average.
- System rarely remains at sustained full capacity.
- Message loss is isolated (minimal clustering) and infrequent.

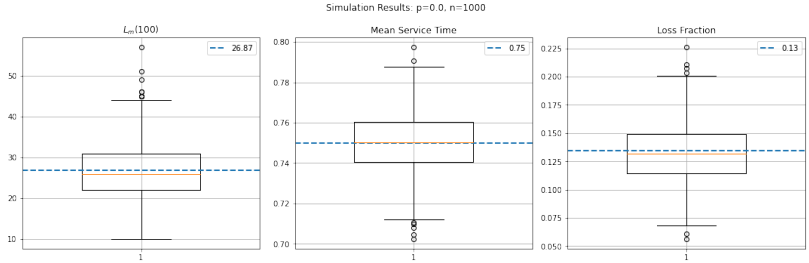
Monte Carlo Results

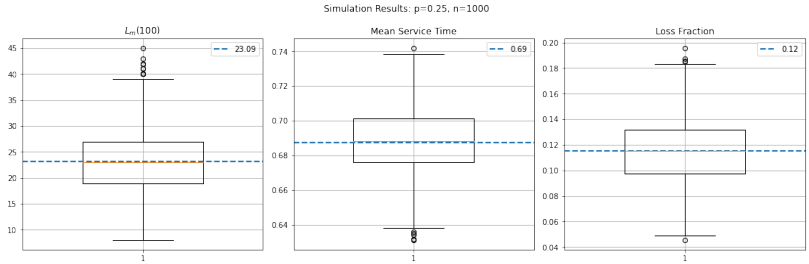
Monte Carlo Estimates vs. Theoretical Benchmarks

	Number Lost			Service Time			Loss Fraction		
p	Mean	Std.	$\mathbb{E}[L_m(100)]$	Mean	Std.	$\mathbb{E}[X]$	Mean	Std.	p_3
0.00	26.69	6.77	26.87	0.750	0.015	0.750	0.132	0.027	0.134
0.25	23.18	6.07	23.09	0.688	0.019	0.688	0.115	0.025	0.115
0.50	19.79	5.58	19.39	0.626	0.021	0.625	0.098	0.023	0.097
0.75	15.72	5.01	15.85	0.563	0.021	0.563	0.078	0.022	0.079
1.00	12.55	4.50	12.50	0.500	0.022	0.500	0.062	0.020	0.063

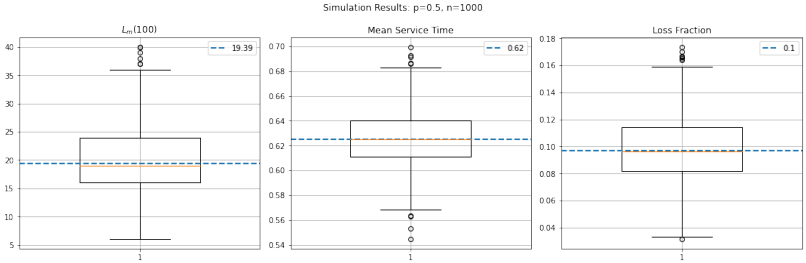
- Sampling distributions constructed from 1000 independent runs per value of p.
- Empirical means closely match theoretical predictions across all metrics.
- Variability decreases as p increases (lower offered load).

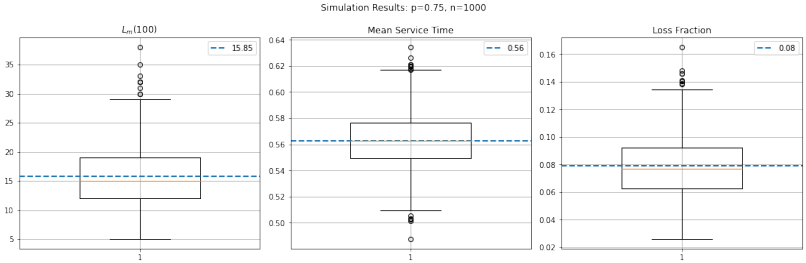
Monte Carlo Validation: $p = 0$ and $p = 0.25$



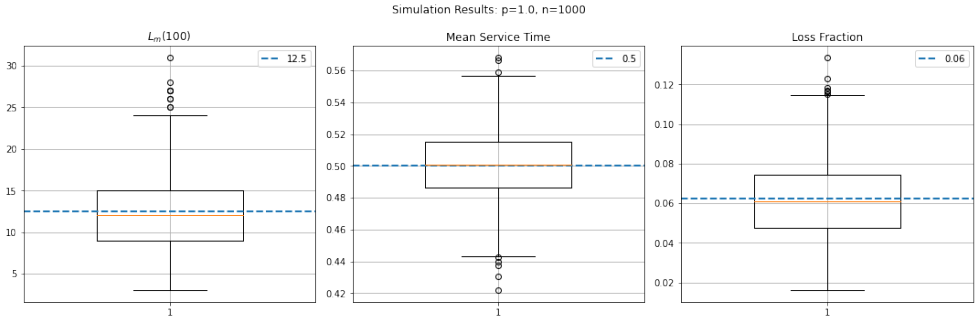


Monte Carlo Validation: $p = 0.5$ and $p = 0.75$





Monte Carlo Validation: $p = 1$



- MC sampling distributions are tightly centered at theoretical values.
- Mean service time simulation results align with analytic values.
- Variability is lowest in the always good weather case ($p = 1$).
- Positive skewness in distribution of number of messages lost and loss fraction (congestion events).

Testing Simulation Results

To formally compare simulation output with analytic predictions for number lost, mean service time, and fraction lost, I perform:

- One-sample t-tests comparing empirical means to theoretical values
- Wilcoxon signed-rank tests as a nonparametric robustness check

across the Bernoulli grid $p \in \{0, 0.25, 0.5, 0.75, 1\}$. I test at α=5% with a Bonferroni correction per metric for multiple testing

$$\alpha_{\text{Bonf}} = \frac{0.05}{5} = 0.01.$$

Results:

- For $L_m(100)$ and $\widehat{\mathbb{E}[X]}$, I fail to reject H_0 for all tests
- For $\widehat{p}_3$, I reject H_0 with both tests for $p = 0$
- The discrepancy at $p = 0$ is small: $0.132108 - 0.134328 = -0.002220$
- Simulation means are statistically indistinguishable from analytic predictions.

Limitations

- Analysis uses steady-state blocking probabilities over a finite horizon.
- Weather is modeled as i.i.d. across arrivals.
- Simplified service time distributions (Uniform and Beta).
- Results rely on Erlang-B insensitivity and model assumptions.
- Simulation estimates subject to finite-sample variability.

Summary

- Model as an $M/G/3/3$ loss system with weather-dependent service times.
- The service time distribution is a Bernoulli mixture with
$$\mathbb{E}[X] = \frac{3-p}{4}.$$
- By the insensitivity property, message loss depends only on the mean service time, so Erlang-B applies with
$$a = \frac{3-p}{2}.$$
- The probability of a lost message is
$$p_3 = \frac{(3-p)^3}{201 - 87p + 15p^2 - p^3},$$
and the expected number of lost messages in 100 hours is
$$\boxed{\mathbb{E}[L_m(100)] = 200\, p_3}.$$
- Simulation results are consistent with the analytic predictions across tested p values.

References I

Willig, A. (1999).
A short introduction to queueing theory.
Technical University Berlin, Telecommunication Networks Group.
Retrieved from https://www.cs.ucf.edu/~lboloni/Teaching/EEL6785_Fall2010/slides/QueueingTheory.pdf

Lockhart, R. (2011).
Queuing theory.
Simon Fraser University, STAT 870 lecture notes.
Retrieved from https://www.sfu.ca/~lockhart/richard/870/11_2/lectures/Queuing/web.pdf

Jain, R. (2012).
Introduction to queueing theory.
Washington University in St. Louis.
Retrieved from https://www.cse.wustl.edu/~jain/queue/ftp/q_30iqt.pdf

Ross, S. M. (2019).
Introduction to probability models (12th ed.).
Academic Press.

Zukerman, M. (2013).
Introduction to queueing theory and stochastic teletraffic models.
arXiv:1307.2968.
Retrieved from https://arxiv.org/abs/1307.2968

References II

Sarkar, J.
Erlang loss formulas: An elementary derivation.
Indiana University–Purdue University Indianapolis.
Retrieved from https://scholarworks.indianapolis.iu.edu/server/api/core/bitstreams/2be23fcc-20b0-4e59-ba65-7ae4d77ec8af/content

Smith, J. M. (2003).
M/G/c/K blocking probability models and system performance.
Performance Evaluation, 52(4), 237–267.
https://doi.org/10.1016/S0166-5316(02)00190-6

Mandjes, M., Taylor, P. G., & De Turck, K. (2017).
The Markov-modulated Erlang loss system.
Performance Evaluation, 116, 53–69.
https://doi.org/10.1016/j.peva.2017.08.005

Cheah, J. Y., & MacGregor Smith, J. (1994).
Generalized M/G/C/C state dependent queueing models and pedestrian traffic flows.
Queueing Systems, 15, 365–386.
https://doi.org/10.1007/BF01189246

Araújo, H. F., de Toledo, M. L. G., Cruz, F. R. B., & MacGregor Smith, J. (2004).
Generalized M/G/c/c state dependent queueing network analysis.
Conference paper.
Retrieved from https://www.est.ufmg.br/ftp/fcruz/publics/cba04.pdf

Appendix: Derivation of $\mathbb{E}[X]$

By the law of iterated expectation,

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid W]\big] = p\,\mathbb{E}[X \mid W = 1] + (1-p)\,\mathbb{E}[X \mid W = 0].$$

Using

$$\mathbb{E}[X \mid W = 1] = \int_0^1 x\,dx = \frac{1}{2},$$

and

$$\mathbb{E}[X \mid W = 0] = \int_0^1 x\,(3x^2)\,dx = 3\int_0^1 x^3\,dx = \frac{3}{4},$$

I obtain

$$\mathbb{E}[X] = p \cdot \frac{1}{2} + (1-p) \cdot \frac{3}{4} = \frac{2p + 3 - 3p}{4} = \boxed{\frac{3-p}{4}}.$$

Appendix: Derivation of $\mathrm{Var}(X)$

First compute

$$\mathbb{E}[X^2] = p\,\mathbb{E}[X^2 \mid W = 1] + (1-p)\,\mathbb{E}[X^2 \mid W = 0].$$

Now

$$\mathbb{E}[X^2 \mid W = 1] = \int_0^1 x^2\,dx = \frac{1}{3}, \qquad \mathbb{E}[X^2 \mid W = 0] = \int_0^1 x^2(3x^2)\,dx = \frac{3}{5}.$$

Thus

$$\mathbb{E}[X^2] = p \cdot \frac{1}{3} + (1-p) \cdot \frac{3}{5} = \frac{9-4p}{15},$$

and

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{9-4p}{15} - \left(\frac{3-p}{4}\right)^2 = \boxed{\frac{3}{80} + \frac{p}{120} - \frac{p^2}{16}}.$$

Appendix: Computing p_3

Using

$$p_3 = \frac{a^3/6}{1 + a + a^2/2 + a^3/6}, \qquad a = \frac{3-p}{2},$$

and substituting a gives

$$\begin{aligned}
p_3 &= \frac{(3-p)^3/48}{1 + (3-p)/2 + (3-p)^2/8 + (3-p)^3/48} \\
&= \frac{(3-p)^3}{48 + 24(3-p) + 6(3-p)^2 + (3-p)^3} \\
&= \frac{(3-p)^3}{48 + 72 - 24p + 54 - 36p + 6p^2 + 27 - 27p + 9p^2 - p^3} \\
&= \boxed{\frac{(3-p)^3}{201 - 87p + 15p^2 - p^3}}.
\end{aligned}$$

Appendix: Full Monte Carlo Summary

Number Lost

p	Count	Mean	Std.	Min	25%	50%	75%	Max	$\mathbb{E}[L_m(100)]$
0.00	1000	26.69	6.77	10	22	26	31	57	26.87
0.25	1000	23.18	6.07	8	19	23	27	45	23.09
0.50	1000	19.79	5.58	6	16	19	24	40	19.39
0.75	1000	15.72	5.01	5	12	15	19	38	15.85
1.00	1000	12.55	4.50	3	9	12	15	31	12.50

Service Time

p	Count	Mean	Std.	Min	25%	50%	75%	Max	$\mathbb{E}[X]$
0.00	1000	0.750	0.015	0.702	0.741	0.750	0.760	0.797	0.750
0.25	1000	0.688	0.019	0.631	0.676	0.688	0.701	0.742	0.688
0.50	1000	0.626	0.021	0.545	0.611	0.625	0.640	0.699	0.625
0.75	1000	0.563	0.021	0.487	0.550	0.563	0.577	0.634	0.563
1.00	1000	0.500	0.022	0.422	0.486	0.501	0.515	0.568	0.500

Loss Fraction

p	Count	Mean	Std.	Min	25%	50%	75%	Max	p_3
0.00	1000	0.132	0.027	0.056	0.115	0.132	0.150	0.226	0.134
0.25	1000	0.115	0.025	0.045	0.097	0.115	0.132	0.196	0.115
0.50	1000	0.098	0.023	0.031	0.082	0.096	0.114	0.174	0.097
0.75	1000	0.078	0.022	0.026	0.063	0.077	0.092	0.165	0.079
1.00	1000	0.062	0.020	0.016	0.048	0.061	0.075	0.134	0.063

Appendix: Test Setup

For each metric and value of p, let $Y_1, \ldots, Y_n$ denote the $n = 1000$ independent simulation replications, with theoretical value $\theta(p)$. The sample mean is

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^{n} Y_i.$$

By the Central Limit Theorem,

$$\frac{\bar{Y} - \theta(p)}{S_Y / \sqrt{n}} \approx \mathcal{N}(0, 1).$$

I perform a one-sample t-test to compare means:

$$H_0 : \mathbb{E}[Y] = \theta(p), \qquad H_A : \mathbb{E}[Y] \neq \theta(p).$$

As a nonparametric robustness check, I apply the Wilcoxon signed-rank test. Let

$$D_i = Y_i - \theta(p).$$

The test evaluates whether the distribution of D_i is centered at zero,

$$H_0 : \text{median}(D_i) = 0, \qquad H_A : \text{median}(D_i) \neq 0.$$

Appendix: Full Statistical Test Results

Variable	p	Theory	Sample	Diff.	T	p_T	p_W	t Conc.	Wil. Conc.
$L_m(100)$	0.00	26.8657	26.6910	-0.1747	-0.8165	0.4144	0.1870	Fail to Reject	Fail to Reject
$L_m(100)$	0.25	23.0856	23.1840	0.0984	0.5129	0.6081	0.2653	Fail to Reject	Fail to Reject
$L_m(100)$	0.50	19.3949	19.7910	0.3961	2.2456	0.0249	0.0547	Fail to Reject	Fail to Reject
$L_m(100)$	0.75	15.8461	15.7210	-0.1251	-0.7901	0.4297	0.1367	Fail to Reject	Fail to Reject
$L_m(100)$	1.00	12.5000	12.5460	0.0460	0.3231	0.7467	0.3445	Fail to Reject	Fail to Reject
$\widehat{\mathbb{E}[X]}$	0.00	0.7500	0.7502	0.0002	0.4058	0.6850	0.4223	Fail to Reject	Fail to Reject
$\widehat{\mathbb{E}[X]}$	0.25	0.6875	0.6883	0.0008	1.4077	0.1595	0.1748	Fail to Reject	Fail to Reject
$\widehat{\mathbb{E}[X]}$	0.50	0.6250	0.6257	0.0007	1.0230	0.3065	0.4022	Fail to Reject	Fail to Reject
$\widehat{\mathbb{E}[X]}$	0.75	0.5625	0.5631	0.0006	0.9704	0.3321	0.4843	Fail to Reject	Fail to Reject
$\widehat{\mathbb{E}[X]}$	1.00	0.5000	0.5004	0.0004	0.5722	0.5673	0.3656	Fail to Reject	Fail to Reject
$\widehat{p}_3$	0.00	0.1343	0.1321	-0.0022	-2.6454	0.0083	0.0037	Reject	Reject
$\widehat{p}_3$	0.25	0.1154	0.1149	-0.0005	-0.6994	0.4845	0.3973	Fail to Reject	Fail to Reject
$\widehat{p}_3$	0.50	0.0970	0.0979	0.0009	1.1819	0.2375	0.5941	Fail to Reject	Fail to Reject
$\widehat{p}_3$	0.75	0.0792	0.0781	-0.0012	-1.6704	0.0952	0.0165	Fail to Reject	Fail to Reject
$\widehat{p}_3$	1.00	0.0625	0.0619	-0.0006	-0.9576	0.3385	0.0812	Fail to Reject	Fail to Reject

p_T is one-sample t-test p-value and p_W is Wilcoxon signed-rank p-value. Reject H_0 for $p < \alpha_{\text{Bonf}} = 0.05/5 = 0.01$.